

Price discovery and information in an emerging market: Evidence from China*

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Abstract

In a developing stock market such as China, where do informed traders locate? Does information originate with traders able to access companies, officials and insiders or are foreigners better informed? We examine the price discovery process for Chinese stocks simultaneously traded in Mainland China and the Hong Kong market. The mainland contributes most to price discovery, according to Hasbrouck's (1995) IS method and the Gonzalo and Granger (1995) PT method. We find a significantly positive relationship between the mainland's share of price discovery and the relative adverse selection component of the effective spread. Consequently, the informational advantage of domestic investors would appear to be the reason that the mainland dominates price discovery.

Key words: Price discovery, Cross-listing, Cointegration, Chinese securities
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1. Introduction

One of the most important functions of the security markets is price discovery – the impounding and reflecting of new information into the equilibrium security prices that are observable to the market. For the same, or intrinsically related, securities traded at multiple markets, the central concern is how the informational price discovery among these trading venues occurs. In particular, who are the most informed traders and what is the nature and source of their informational advantage? Once we know which venue dominates the price discovery process, it is possible to infer answers to some of these more fundamental questions.

The purpose of this paper is to analyze the price discovery process of the Chinese A- and H- shares. These are stocks issued by the same Chinese companies but traded respectively in the Mainland China (either Shanghai Stock Exchange or Shenzhen Stock Exchange) and Hong Kong stock markets. We show that the financial development of Mainland China's emerging market has a significant impact on the price discovery process. Moreover, using a sizeable period of intraday data (year by year for seven consecutive years), our results indicate an important time-varying characteristic of cointegration between A- and H- share prices. We show that cointegration of a trading pair of stocks is dependent on a sufficient number (or volume) of Hong Kong trades relative to the corresponding number (or volume) of Mainland China trades. The empirical results also suggest that the Chinese mainland market contributes more to the price discovery process than the Hong Kong market for most of the A-H pair stocks, in most years. Our findings thus support the home-bias hypothesis for cross-listing stock trading. In other words, price discovery will take place in the domestic market since investors have more information about domestic stocks.

Importantly and not attempted by any other study we are aware of, we find that the adverse selection component of bid-ask spread in the A-shares market is significantly positively related to its contribution to price discovery relative to the H-shares market. This finding implies an information advantage for domestic investors in the Mainland China market. Further, it is supportive of the hypothesis that the well-established home bias of investors towards domestic stocks is due to geography and proximity. The far stronger price discovery in the local market reflects traders with the greatest contact with and proximity to informed insiders that ensures a powerful informational

advantage. These insiders could be company officials, bureaucrats or party officials in influential positions.

In previous research, several econometric approaches have been applied to intraday data of securities simultaneously traded at different security exchanges (even in different countries), in order to measure the contributions of price discovery for each trading venue. We divide the literature into two categories according to their econometric methods and measures of contribution: the first category is Hasbrouck's (1995) Information Share (IS) and the second, Gonzalo and Granger's (1995) Permanent-Transitory decomposition (PT) method. Both econometric approaches seek to provide an explicit measure of each market's contribution to price discovery.

The first category assumes an implicit unobservable efficient price common to all markets, while building on earlier contributions, Hasbrouck (1991a, b). The information share of a particular market is the proportional contribution of that market's innovations to the innovation in the common efficient price. Applications (or adjusted applications) are found in Booth et al. (2002), Chan et al. (2002), Hupperets and Menkveld (2002), Eun and Sabherwal (2003), Hasbrouck (2003) and Chakravarty et al. (2004), to name but a few.

The second category is the common factor method characterized by Gonzalo and Granger (1995)'s PT decomposition of cointegrated time series and is applied in Harris et al. (1995, 2002a), Booth et al. (1999), Booth et al. (2002) and Chakravarty et al. (2004). It defines each market's contribution to price discovery as a proportional function of its error-correction adjustment coefficients.

Both the IS and the PT methods are based on a vector error correction model (VECM) but use different definitions of contribution to price discovery and decomposition techniques.¹ The main findings are that the two models provide similar results if the residuals are uncorrelated between markets, but typically different results when substantial correlation exists.

Additionally, several approaches examine the way in which security prices incorporate information. These include bid-ask spread decompositions (e.g., Lin et al., 1995, Huang and Stoll, 1997), structural model of intraday price formation (e.g.,

¹ A special issue of *Journal of Financial Markets* has a discussion of the similarity and differences between both methods, including arguments put forward by the initiators. Detailed econometric analysis, comparison of simulation results and theoretical explanation in this special issue include the papers of de Jong (2002), Baillie et al. (2002), Lehmann (2002), Harris et al. (2002b), and Hasbrouck (2002).

Madhavan et al., 1997), and microstructure effects on price, such as impacts of block trading (e.g., Kraus and Stoll, 1972).

Analyses of price discovery for cross-listing stocks in different countries utilize these econometric and microstructure methods. The conclusions vary for different regions and stock exchanges. We make a concise summary of this literature in Table 1, presenting their cross-listing countries, methodologies, and main results. Most of these studies focus on the stock markets for two different countries, typically one of which is the U.S., and investigate the pricing process for dually listed stocks. Neither the home market, nor foreign market dominates the price discovery process.

(Insert Table 1 about here)

The literature divides into the following categories:

1). The first category indicates that the dominant price discovery location varies cross-sectionally for different stocks (domination varies). Werner and Kleidon (1996) analyze intraday patterns for U.K. and U.S. trading of British cross-listed stocks. The two-hour overlap is characterized by concentrated trading, as private information originating in New York gets incorporated into prices in both markets. Hupperets and Menkveld (2002) study Dutch stocks cross-listed at NYSE during the one overlapping trading hour. They find no consistent source of price discovery, either Amsterdam or New York. Eun and Sabherwal (2003) examine U.S.-listed Canadian stocks. They find the U.S. share of price discovery ranging from as little as 0.2% to 98.2%, with an average of 38.1%. This share directly relates to the U.S. trading share and inversely to U.S. bid-ask spreads. Wang and Jiang (2003) use daily data to find that H-shares exhibit significant exposure to Hong Kong market factors and behave more like Hong Kong stocks than mainland Chinese stocks. Unfortunately, the data frequency is too coarse to draw strong conclusions. Halling et al. (2007) investigate the location of stock trading for firms with a U.S. cross listing. They find that the fraction of trading that occurs in the United States tends to be larger for companies from countries that are geographically close to the U.S., have low financial development and poor insider trading protection.

2). The second category (home dominates) explicitly presents empirical results for the home-market dominated price discovery process. Biais and Martinez (2004) study opening prices set simultaneously for German-French dual-listing stocks in Frankfurt and Paris. They find that home prices are informationally more efficient than foreign prices. Grammig, Melvin, and Schlag (2005) analyze exchange rates along with

equity quotes for three German firms listed in New York (NYSE) and Frankfurt (XETRA) during overlapping trading hours. They find that most (but not all) of the fundamental or random walk component of firm value is determined in Frankfurt. Pascual, Fuster, and Climent (2006) introduce a methodology that distinguishes trade-related and trade-unrelated informative shocks. They provide evidence that the NYSE does not contribute to price discovery for Spanish cross-listed stocks during the daily (two-hour) overlapping interval. Menkveld, Koopman, and Lucas (2007) use a state space model to study 24-hour (round-the-clock) price discovery for Dutch stocks cross-listed in the U.S. market and suggest a minor role for the NYSE in price discovery for Dutch shares. This is despite the NYSE's non-trivial and growing market share.

Lieberman et al. (1999) for Israel-US cross-listing stocks, Ding et al. (1999) for Malaysia-Singapore cross-listing stocks, Kadapakkam et al. (2003) for India-UK cross-listing stocks, and Chong and Su (2006)¹ on Chinese cross-listing A- and H-shares analyze a developing and mature market. Domowitz, Glen and Madhavan (1998) show the impact of Mexican stocks cross-listed on U.S. stock exchanges but do not analyze which market dominates price discovery. Furthermore, for Chinese securities markets, Chan, Menkveld, and Yang (2007, 2008) examine the price informativeness of Chinese A- and B- shares.² They also examine the relationship between information asymmetry and B-share price discount using relatively old and short-period data around 2001. The main result of these emerging-developed markets cross-listing researches is that the emerging market contributes to most of the price discovery (except the Indian case). This indicates our hypothesis, that the emerging market has an informational advantage, prevails in most of the cases. There is, however, a proviso that the emerging market represents the stock's home base.

¹ They use data of 5 months in 2003 for cross-listing Chinese A- and H- shares to study price discovery, while our intraday data covers seven years and the results show an evolutionary cointegration process for A- and H- share prices. Moreover, different from their explanations, we explain the reason for Mainland China's leading role in price discovery by decomposing the bid-ask spread and examining the information advantage of domestic investors. We also test for the influence of government policies (QFII and QDII) and stock exchanges, and provide significant regression results.

² A-shares and B-shares are stocks issued by the same Chinese company, respectively to domestic investors and foreign investors before Feb 19, 2001. After Feb 19, 2001, the Chinese government adopted a new policy to allow domestic investors with foreign exchanges holdings to buy B-shares. However, there has been a consistent B-share price discount compared with A-share price even after the Feb 19, 2001 policy. Since A- and B- shares are both listed in Mainland China, they are not strictly cross-listing stocks in terms of trading venues.

From a broader perspective, most of these studies merely examine price discovery utilizing a relatively short, once off, dataset. What they neglect are changes taking place over the longer term. In contradistinction, our paper is the first systematic in-depth study of price discovery for stocks cross-listed between an emerging and developed market using intraday data over seven consecutive years. Furthermore, with most previous cross-listing stock studies utilizing U.S. data, the order flow in the cross-listed market is negligible relative to the main market. By contrast, our cross-listing sample has significant order flow for both markets (Mainland China and Hong Kong).

Our study carries out a more intensive and systematic analysis than does any other in this field. We use a unique intraday sample covering seven years to study the Chinese emerging stock market in comparison with the developed Hong Kong market. By applying the Johansen (1991) cointegration test, the Hasbrouck (1995) IS method, and the Gonzalo-Granger (1995) PT decomposition method we analyze each cointegrated year for each stock A-H pair. Our results demonstrate the evolution of emerging market's contribution to price discovery for cross-listed A- and H- shares. We then combine our econometric methods with an important bid-ask spread decomposition from the market microstructure literature. We merge up until now two distinct literatures, so as to investigate the driving force behind our empirical results.

This paper is organized as follows: Section 2 describes the model setting and econometric methods used – Johansen's (1991) cointegration test, Hasbrouck's (1995) IS, and Gonzalo-Granger's (1995) PT decomposition. Section 3 introduces the trading mechanism and intraday data of our Mainland China A-shares and Hong Kong H-shares markets. We present detailed empirical applications of the econometric methods and analysis of results in Section 4. Section 5 decomposes the bid-ask spread to explain the Hasbrouck (1995) IS in terms of fundamentals. Section 6 concludes with a brief summary of our study and results.

2. Model and Methods

In this section, we demonstrate use of the cointegration and error correction models. We show the differences in interpretation of the IS method and PT methods for each market's contribution to price discovery. Consider a pair of stocks for the same company trading at two different venues. Since these assets share a common future cash flow and other common company features such as risk, we expect there to be a

strong arbitrage force closely linking their prices¹. We model their prices to discover the common long-term component that binds them together.

2.1 Cointegration model

Cointegration is a linear dynamic model used to describe the long-run relationship between non-stationary time series data. Let the (log) price vector $p_t = (p_{ct}, p_{ht})'$ denote prices at time t for two markets, where p_{ct} is the price for Mainland China stock market and p_{ht} the price for Hong Kong stock market, both at time t . In our market microstructure applications, we assume the stock prices follow a vector autoregressive process and contain an independently and identically distributed random walk component that represents the random arrival of information. As a result, each price series of two stock markets are non-stationary and integrated of order 1, as denoted $I(1)$. The stock returns Δp_t (log price changes) take the form of a covariance stationary time series process of $I(0)$. We express the implied vector moving average (VMA) form of the prices as:

$$\Delta p_t = \Psi(L)\varepsilon_t = \varepsilon_t + \Psi_1\varepsilon_{t-1} + \Psi_2\varepsilon_{t-2} + \dots, \quad (1)$$

where ε_t is a zero-mean 2×1 vector of serially uncorrelated disturbances with covariance matrix of Ω , satisfying:

$$E[\varepsilon_t \varepsilon_s'] = \begin{cases} 0 & \text{if } t \neq s; \\ \Omega & \text{otherwise} \end{cases},$$

L is the lag operator and Ψ is a polynomial in the lag operator. Since the price differences are stationary, the prices vector process is cointegrated of order 1, as $CI(1,1)$ with a cointegrating vector $\beta = (1, -1)'$, satisfying the condition that $\beta' p_t$ is $I(0)$.

This implies that $\beta' \Psi(1) = 0$, where $\Psi(1) = I + \Psi_1 + \Psi_2 + \dots$, is the sum of the moving average coefficients and has rank 1. According to our assumption with respect to β , the two rows of $\Psi(1)$ are identical and the long-run impacts of an innovation ε_t on each of the prices are the same, represented by $\Psi(1)\varepsilon_t$. In our two-market case, we

¹ Although there is large price discount between Chinese A- and H- shares, our cointegration test show that for certain time periods for these stocks, the arbitrage force links their prices in spite of transaction costs and other impediments. Chen and Swan (2007) explain the sizeable price differences between otherwise identical Chinese A- and B- shares in terms of a liquidity asset pricing model.

denote $\psi = (\psi_c, \psi_h)$ as the common row vector of $\Psi(1)$. As suggested by Stock and Watson (1988), the cointegrated price process has the following common trends representation:

$$p_t = p_0 + \psi \left(\sum_{s=1}^t \varepsilon_s \right) I_2 + \Psi^*(L) \varepsilon_t. \quad (2)$$

This representation of p_t consists of three parts: a 2-vector of initial price values which reflect the non-stochastic differences between two prices, a random walk component fundamentally common to two prices changes, and a zero-mean covariance stationary process.

We use the Johansen (1991) cointegration test in Section 4 to estimate and test for the cointegration relationship between pairs of A- and H-shares prices year by year from 2001 to 2007.

2.2 Information Share (IS) method

According to the price representation of (2), when new information arrives, the increment $\psi \varepsilon_t$ is the component of permanently impounded price change, with its variance $\psi \Omega \psi'$. If Ω is diagonal, which implies that the innovation terms of each market are not correlated, we can determine each market's contribution to the innovation of the random-walk component $\psi \varepsilon_t$. Hence, we define the IS of each market j (c for Mainland China and h for Hong Kong) as the proportion of its innovation to the total variance:

$$IS_j = \frac{\psi_j^2 \Omega_{jj}}{\psi \Omega \psi'} = \frac{\psi_j^2 \omega_j^2}{\psi_c^2 \omega_c^2 + \psi_h^2 \omega_h^2}, \quad j = c, h, \quad (3)$$

where ψ_j is the j^{th} element of ψ and ω_j^2 the j^{th} diagonal element in Ω . By normalization, the sum of the information share $IS_c + IS_h$ is equal to 1, and a higher (lower) IS for market j indicates more (less) incorporation of innovation from new arrival of information.

However, the above representation comes from the assumption that there is no correlation between the price innovations across the two markets. If the covariance matrix Ω is non-diagonal, we follow Hasbrouck (1995) and use the Cholesky factorization to triangularize the covariance matrix Ω . Letting F denote the lower

triangular Cholesky factorization of Ω such that $FF' = \Omega$, the IS of j^{th} (c for Mainland China and h for Hong Kong) market can be redefined as

$$IS_j = \frac{([\psi F]_j)^2}{\psi' \Omega \psi}, \quad j = c, h, \quad (4)$$

where $[\psi F]_j$ is the j^{th} element of the row matrix ψF . Since the factorization imposes a maximized (minimized) information share on the first (last) price, we can calculate the upper (lower) bound of the j^{th} market's information share by permuting ψ and Ω to place that market's price first (last). In our two-market case, we obtain the upper and lower bounds for Mainland China and Hong Kong market respectively. We report the mid-point of the upper and lower boundaries of each market's IS.

2.3 Permanent-Transitory (PT) method

According to Gonzalo and Granger (1995), since both price series are integrated of order 1, $I(1)$, under certain conditions, the price vector can be decomposed to the following PT form:

$$p_t = \Gamma f_t + Az_t = \Gamma \gamma'_{\perp} p_t + A\alpha' p_t, \quad (5)$$

where f_t represents the permanent common component and z_t the transitory component. Γ and A are, respectively, the loading matrix for the permanent component coefficient and transitory component. By construction, the permanent factor f_t is $I(1)$ and the error correction term z_t is stationary, as $I(0)$. The error correction term does not Granger-cause the permanent factor in the long run. Gonzalo and Granger (1995) show that under these assumptions and restrictions, the matrices $\Gamma = \alpha_{\perp}(\gamma'_{\perp} \alpha_{\perp})^{-1}$ and $A = \gamma(\alpha' \gamma)^{-1}$, where $\gamma = (\alpha'_{\perp} \beta_{\perp})^{-1} \alpha'_{\perp}$. By definition, the vector of common permanent factor weights $\gamma_{\perp}$ is orthogonal to the coefficient vector γ such that $\gamma'_{\perp} \gamma = 0$.

We construct the error correction term such that $\beta = (1, -1)'$, so the permanent component Γf_t is a weighted average of observed prices. Therefore, the PT decomposition suggests using each price's weight of the permanent component as its share of price discovery (c for Mainland China and h for Hong Kong):

$$PT_j = \frac{\alpha_{\perp,j}}{\alpha_{\perp,c} + \alpha_{\perp,h}}, \quad j = c, h. \quad (6)$$

In addition, since we have $\alpha'_{\perp} \alpha = 0$, we use the coefficient of the error correction term $\alpha = (\alpha_c, \alpha_h)'$ to express the price discovery share in (6), as

$$PT_c = \frac{\alpha_h}{\alpha_h - \alpha_c}, \quad PT_h = \frac{-\alpha_c}{\alpha_h - \alpha_c}.$$

2.4 Comparison of the IS and PT methods

It is important to point out that although both IS and PT methods try to measure the contribution to price discovery for cross-listing securities, their results may not be exactly the same, because their definition of share of price discovery vary from each other. The special issue of *Journal of Financial Markets* in 2002 contains a detailed discussion of these methods, as already noted. Neither of these two methods attempts to determine which market has the ‘best’ price. They use different measurements to investigate which market moves first in reaction to new information. Additionally, which market is more efficient in incorporating new information? The IS method emphasizes each market’s contribution to new information innovation and the PT method emphasizes the adjustment of each market price to their long-run cointegrated relationship.

3. Data Description

In this section, we briefly introduce the history and relationship between stock exchanges of Mainland China and Hong Kong. We also compare the trading hours and features of both markets. We present summary statistics of the cross-listing companies and discuss details of the sample intraday data.

3.1 Two markets and cross-listing

We analyze the price discovery process for stocks of 30 Chinese companies cross-listed in the Mainland China and Hong Kong stock markets, respectively as A- and H-shares. There are two stock exchanges in Mainland China: the Shanghai Stock Exchange (SSE) established in November 1990 and the Shenzhen Stock Exchange (SZSE) established in July 1991. While in Hong Kong (Special Administrative Region of China), in March 1999, under the comprehensive market reform of the stock and futures markets, the Stock Exchange of Hong Kong Limited (SEHK), Hong

Kong Futures Exchange Limited (HKFE) and Hong Kong Securities Clearing Company Limited (HKSCC) merged and became wholly-owned subsidiaries of a single holding company, HKEx. Companies incorporated in China can issue A-shares to its domestic investors and trade in Chinese currency Yuan, listed either in SSE or in SZSE. With the approval of the China Securities Regulatory Commission (CSRC), these companies can also issues H-shares to international investors in Hong Kong stock market, traded in Hong Kong dollars and listed in HKEx.

On 19 June 1993, the SEHK, the CSRC, the SSE and the SZSE signed a Memorandum of Regulatory Co-operation in Beijing, which paved the way for the listing of mainland-registered enterprises in Hong Kong. On 15 July 1993, Tsingtao Brewery Company Limited became the first mainland enterprise to list its H-shares on the Stock Exchange. Most of the cross-listing companies firstly issued H-shares in Hong Kong and return to Mainland China to issue A-shares and thus become cross-listing stocks in these two markets. In recent years, there have been more and more companies simultaneously cross-listed in Mainland China and Hong Kong. Until June 3, 2008, there were 60 pairs of cross-listing A- and H- shares. The reason why we only choose 30 cross-listing pairs is that all these 30 pairs cross-listed before or in the year 2006 so that we have at least two years (our sample covers intraday data from the year 2001 to 2007) to analyze. We eliminate other cross-listing stocks from our sample because of their relatively short listing period or incompleteness with respect to intraday data.

The trading hours and trading mechanisms are the same for SSE and SZSE but differ between SSE (SZSE) and HKEx. They all conduct trading from Monday to Friday except for public holidays that could differ. Figure 1 displays for each trading day the overlapping and non-overlapping trading hours for the Mainland China stock market and the Hong Kong stock market. There are three trading periods for both markets: the pre-open period with a call auction, the morning session with a continuous auction, and after a lunch break, the afternoon session with a continuous auction. For our purpose, we choose the overlapping trading periods of each morning session and afternoon session for both markets with continuous auctions. Those periods range from 10:00 ~ 11:30 in the morning and from 14:30 ~ 15:00 in the afternoon.

(Insert Figure 1 about here)

In addition, our sample eliminates days when only one market has data, since our cointegration and price discovery process analysis requires interaction between two markets.

3.2 Data sources and sample details

We construct our sample of A- and H- shares intraday data from the SIRCA (Securities Industry Research Centre of Asia-Pacific) Reuters database. It provides the quote and transaction intraday data for stocks traded in Mainland China and Hong Kong stock markets, sorted by symbol (company name). As we have explained, we choose 30 A- and H- shares pairs which cross-listed before or in 2006 with no missing observations. In order to present the time-varying characteristic of our result and the development of the emerging market, our intraday sample covers seven years, from 2001 to 2007. Table 2 shows the pairs of cross-listing stocks, with the company names, A- and H- share codes and their date of IPOs on each market:

(Insert Table 2 about here)

Since most stocks do not have intraday data in Mainland China until 2003 for technological reasons, we can only cover the earliest possible date for them during the period from 2001 to 2007. Our final sample consists of 26 companies cross-listed in SSE and HKEx, and four companies cross-listed in SZSE and HKEx. From first-listing locations, we can clearly see that most of the sample companies (28 out of 30) firstly had IPOs in Hong Kong stock market, only ZTE and CHINA MERCHANT BANK firstly had IPOs in Mainland China. This directly relates to the policy direction of the Chinese government in the 1990s.

We could base analysis of cross-listing stocks on either transaction or quoted spread prices. Since the use of transaction prices could suffer from bid-ask bounce, we adopt the mid-quote price sampling method to perform our analysis. In particular, we select the midpoint of the bid and ask quotes in each market at 1-minute fixed time intervals.¹ If there are no quotes at exactly the 1-minute time point, we use the bid-ask quotes for the closest happened transaction to each time point. Consequently, for each stock in each market, we have 122 prices for analysis per day (10:00 ~ 11:30 in the morning and 14:30 ~ 15:00 in the afternoon). Since A-shares in SSE (SZSE) trade in Chinese *Yuan* and H-shares in HKEx trade in *Hong Kong dollars*, we use the

¹For robustness and consistency, we also perform the analysis by using 5-minute and 10-minute intervals and there is no qualitative change in results.

corresponding days' exchange rates between *Yuan* and *Hong Kong dollars* to convert the prices of H-shares to comparable Chinese *Yuan* to make the prices consistent in terms of denomination and magnitude. We obtain from Datastream the daily exchange rate between *Chinese Yuan* and *Hong Kong dollars*.

4. Empirical Results and Analysis

4.1 Results of cointegration test

As described in the data section, we use one-minute-interval; mid-quote time-series intraday prices for pairs of A-share and H-share in each year to estimate the Johansen (1991) cointegration test statistics. We present estimation and results in Table 3, with the asterisk implying a cointegration relationship at a 5% significant level between A- and H- shares prices for a given stock pair in a given year. The last four rows in Table 3 respectively shows the number of cointegrated A-H pairs, the number of total A-H pairs, the frequency of significant A-H cointegrated pairs and the percentage of significant A-H cointegrated pairs in each year from 2001 to 2007.

(Insert Table 3 about here)

To what extent did the gradual opening up of China's domestic capital market contribute to cointegration? The most important reforms are the QFII and QDII policies adopted by Mainland China over this period. Qualified Foreign Institutional Investor (QFII) is a scheme that allows qualified foreign financial institutions to trade Chinese A-shares via special accounts opened at designated custodian banks. Similarly, Qualified Domestic Institutional Investor (QDII) is a capital market scheme set up to allow financial institutions in Mainland China to invest in offshore markets such as securities and bonds in Hong Kong. These are both transitional arrangements that provide limited opportunities for domestic (foreign) investors to access foreign (domestic) markets. Capital is unable be transferred freely in and out of the country since China's currency is neither traded nor freely floated. The China Securities Regulatory Commission (CSRC) and the People's Bank of China (PBoC) introduce the QFII program on November 5, 2002 to allow foreign capital to access China's financial markets. The Chinese Government announces the QDII scheme on 13 April 2006, allowing Chinese institutions and residents to entrust Chinese commercial banks to invest in financial products overseas. These are limited to fixed-income and money market products. Furthermore, the Chinese government announced on 11 May

2007 a scheme to widen the scope of the QDII investment so that banks can invest in stocks-related products, but with certain restrictions. This deregulation of investment between Mainland China and Hong Kong since 2002 provides more opportunities for financial institutions to trade simultaneously in both markets and to seek arbitrage opportunities. Especially after the introduction of a policy to allow Mainland banks to invest in stocks in Hong Kong market in 2007, we observe a rapid increase in cointegrated A-H pairs.

Based on the trend towards cointegration for A- and H-shares and capital market policy changes, we investigate the reason for A-H pair cointegration. We hypothesize that relative trading activity, such as trading frequency, trading volume, and transaction costs, influence whether A- and H-shares cointegrate. We use the indicator variable, *COINT*, (*COINT* = 1, otherwise *COINT* = 0) in conjunction with Logistic regression analysis to indicate if a pair of stocks is cointegrated in a given year. We investigate three independent variables, *R_count*, *R_vol_sum* and *R_spread*. They are, respectively, the ratio of trading frequency of Mainland China to that of Hong Kong, the ratio of total trading volume of Mainland China to that of Hong Kong, and the ratio of relative spread of Mainland China to that of Hong Kong, for a given A-H pair in a given year. From the Logistic regression, we do find a very significant negative relationship between *COINT* and *R_count* (alternatively, *R_vol_sum*), but no significant relationship between *COINT* and *R_spread*. We report the results for our regression in Table 4.

(Insert Table 4 about here)

The Logistic regression results in Table 4 indicate that an increase in trading activity in Hong Kong, either in the number or volume of trades, relative to Mainland China leads to more significant cointegrational relationships between A-and H- shares. We were unable to detect any threshold number or volume of Hong Kong trades. It is consistent with our expectation that the capital market policy changes and gradual opening up of offshore investment by Mainland China contribute to the gradual cointegration between and A- and H- shares market prices. However, we stress that cointegration does not necessarily mean a reduction in or disappearance of the price premium of MC over HK stocks.

Next, we report the empirical results for IS method and PT method. Since both models base their analysis on the vector error correction model (VECM) previously

that requires cointegration the two time-series of intraday prices, we select just the cointegrated years for these 30 stocks and apply both methods to the data subset.

4.2 Empirical results of Information Share (IS) method

According to the Cholesky factorization described in the method section, we obtain the upper and lower bounds of IS for each stock in each market. Our results show that the upper and lower bounds are not widely separated from each other (within 0.2), which indicates that the correlation of innovations between the two markets is not large. Therefore, for simplicity and to provide for easy comparisons, we report the mid-point of the upper and lower bounds of IS of Mainland China and Hong Kong stock markets in Table 5. For each stock pair in each cointegrated year, the IS of MC and HK sum up to one, according to our methodology. Consequently, an IS higher than 0.5 in a certain market indicates that this market's contribution to innovation and price discovery is larger than the other market, in terms of the IS explanation.

(Insert Table 5 about here)

From Table 5, it is clear that, for almost all pairs of A-H stocks in their cointegrated years, the Mainland China market's ISs are higher than 0.5. According to our IS model, a higher (lower) IS for a certain market indicates its ability to contribute more (less) to the innovation from new arrival of information. Therefore, our empirical results provide strong evidence that the Mainland China market plays a predominant role in price discovery for cross-listed A- and H- shares. Although the IS for Mainland China market ranges from 0.496 to 0.950 and the IS for Hong Kong market ranges from 0.050 to 0.505, of all the 77 cointegrated pairs used in this sample, only six ISs for Mainland China are higher than 0.7. This indicates that, although Mainland China market dominates the price discovery for cross-listing A- and H- shares, its relative informativeness is only slightly larger than the informativeness of Hong Kong's market.

By contrast, we find for Bank of China (BOC) whose A-shares were listed on July 5, 2006 and H-shares on June 1, 2006, the IS in Hong Kong stock market is slightly larger than for Mainland China. A possible explanation is that Bank of China in Hong Kong was established and incorporated some years early in 2001. It has been a very important part of BOC, since historically it has been a commercial bank focusing on overseas business. Therefore, it is reasonable to believe that investors in BOC in Hong Kong have an informational advantage relative to Mainland China investors.

To illustrate Table 5 more directly, we present in Figure 3 the distribution and accumulated distribution of Information Shares for Mainland China in all the cointegrated years of our sample A- and H- shares, according to the estimation of Table 3. The dark bars show the distribution of ISs for Mainland China and the light area shows the accumulated distribution of ISs for Mainland China. We observe a concentrated distribution of ISs for Mainland China market of between 0.5 and 0.7, that indicates its dominant role in price discovery.

A natural question to ask at this stage is: why the Mainland China market has a leading role in price discovery and why the trading in Mainland China is more informative than the Hong Kong market for cross-listing A- and H- shares? We will explore the reasons after presenting the results for PT method in the following section.

(Insert Figure 3 about here)

4.3 Empirical results of Permanent-Transitory (PT) method

As we have explained in the model section, the PT decomposition suggests using each price's permanent component weight as its share of price discovery. We apply this method to the A-H pairs in their cointegrated years and obtain empirical results for shares of price discovery respectively for the Mainland China and Hong Kong markets. Because of normalization, the shares of price discovery for both markets for the same pair in a certain cointegrated year sum up to one. Therefore, the same as for the IS method, a share of price discovery for a certain market higher than 0.5 indicates a leading role in price discovery for this market. We bold the shares of price discovery for Hong Kong market when they are higher than 0.5, which means that in these years and for these A-H pairs, the Hong Kong market plays a leading role in price adjustment.

(Insert Table 6 about here)

From Table 6, it is not clear which market predominantly leads price discovery and price adjustment. For the A-H pairs in their cointegrated years, 21 out of 77 results imply that the Hong Kong market plays a leading role in price discovery in terms of PT method. For the remaining 56 results, the Mainland China market has a share of price discovery higher than 0.5 and leads the price adjustment for cross-listing A- and H- shares. To be clearer, in Figure 4, we present the distribution and accumulated distribution of price discovery shares for Mainland China using the PT method in all the cointegrated years for our sample A- and H- shares. These correspond to estimates

presented in Table 3. The dark bars show the distribution of shares of price discovery for Mainland China and the light area shows the accumulated distribution of shares of price discovery for Mainland China. There is a dense distribution of shares of price discovery for Mainland China with value higher than 0.5, which indicates its leading role in price discovery. Nevertheless, there are also a few sample pairs in certain years, in which the H-shares market plays a leading role in price discovery.

(Insert Figure 4 about here)

To illustrate the empirical results from Figure 4 and Table 6, we cannot justify saying that the Mainland China market significantly dominates price discovery for all sample pairs in their cointegrated years. However, 72.7% (56/77) of the results do indicate that Mainland China market's share of price discovery is higher than that of the Hong Kong market and that Mainland China market leads price adjustment for A-H pairs in these years.

4.4 Comparison of the results from two methods

Although the results from the two methods, respectively IS and PT, are not exactly the same as each other, it is reasonable to draw the conclusion that Mainland China market dominates the price discovery for most cross-listing A- and H- shares during their cointegrated years. Notice that in Section 2.4, we have compared the differences between these two methods in terms of definition and measurement of shares of price discovery. The IS method focuses on the contribution of each market to the innovation of new arrival of information, while the PT method focuses on the adjustment of price series to fit the long-run relationship between two cointegrated series. Therefore, in terms of 'variance' measures, namely the IS, Mainland China market contributes significantly more to price discovery. By comparison, this advantage of Mainland China market is relatively weaker in terms of price adjustment, because results from the PT method show that the Hong Kong market also leads Mainland China for quite a few A-H pairs in certain cointegrated years.

What caused the disparity of shares of price discovery between cross-listing A- and H- shares? We tackle this question in Section 5, by combining our empirical results with market microstructure features in reasoning order to discover why the A-shares market dominates most of the price discovery.

5. The explanation for IS and PT results

Since IS and PT¹ measure the contribution to price discovery, we set out to explain the cross-section and time-series differences of ISs and PTs for our sample A- and H-shares in their cointegrated years.

We hypothesize that Mainland China will have a higher IS or PT because the domestic investors in Mainland China have an information advantage and geographic proximity compared to foreign investors in the Hong Kong stock market. We also expect the various policy changes relating to China's securities regulation to have an influence on the domestic investors' position of informational advantage, hence changing the relative IS of Mainland China.

The literature commencing with French and Poterba (1991) that tracks the information disadvantage of foreign investors, terms this issue the home bias puzzle. It shows that due to an information disadvantage, foreign investors tend to underweight their international component of portfolios. Cooper and Kaplanis (1994) and Tesar and Werner (1995) investigate possible explanations for home bias in different situations. They indicate an information advantage as a potential explanation. In terms of geography proximity for domestic investors (see, for example, Grubel (1968), Levy and Sarnat (1970) and Solnik (1974), and Coval and Moskowitz (1999, 2001)), a possible explanation is that investors geographically in proximity to the firm's headquarters gain an informational advantage. Feng and Seasholes (2004) find highly correlated purchasing and sale behavior for geographically categorized investors in Mainland China. Furthermore, investors living near a firm's headquarters react in a similar manner to releases of public information, indicating that geography proximity is an important factor for the behavior of informed investors. Hong, Kubik, and Stein (2005) find that proximity and word-of-mouth effects are important in the transmission of information. A natural question to pose is what are the consequences for stock returns if geographic proximity indicates a higher proportion of private relative to public information contained in the stock price? Easley and O'Hara (2004) show that within the context of an asset-pricing model, such compositional differences can affect the cost of capital, with investors demanding a higher return to hold stocks with greater private information.

Studies particularly focusing on the relationship between investment performance and information advantage (or disadvantage) of domestic (or foreign) investors,

¹ We use PT to represent the share of price discovery using PT method, for the rest of the paper.

especially in emerging markets, can be divided into two categories. The first category shows that domestic investors have better investment performance than foreign investors, but the reason varies according to different countries and samples. Both Dvořák (2005) and Agarwal et al. (2009) focus on the emerging Indonesia stock markets. Dvořák (2005) proposes the information advantage hypothesis of domestic investors, suggesting that the combination of local information and global expertise leads to higher profits. While Agarwal et al. (2009) explain the general underperformance of foreign investors by their aggressive trading behavior. Choe et al. (2005) document the trade-weighted disadvantage of foreign money managers in the developing Korean market, compared with domestic ones, contributing it to the poor timing of trading by foreign investors. Grinblatt and Keloharju (2001) document that investors are more likely to trade stocks of Finnish firms that have close distance, same language and similar culture with themselves. Ivković and Weisbenner (2005) find a strong preference for local investments by households and suggesting that they can exploit local information. Few studies fall into the second category, which documents underperformance of domestic investors. For example, Chen et al. (2009) shows that foreign investors outperform domestic investors during the trading behavior ahead of open-ending events of Taiwanese closed-end funds, holding constant access to information.

In our paper, we use panel regression analysis to test the hypothesis that the informational advantage of Mainland China domestic investors, represented by the adverse selection component of the bid-ask spread, is reflected in the stronger price discovery measure obtained by Hasbrouck's (1995) IS and Gonzalo-Granger's (1995) PT. We also test the cross-sectional effects of industries, stock exchanges, and the time-series effects of policy changes on the IS and PT. We firstly discuss the dependent variable and the reasons of choosing the explanatory variables, together with the proposed hypothesis. Then, we present and discuss the panel-data regression results and their implications.

5.1 Dependent variable

The dependent variable is the Logistic transformation of the IS (Hasbrouck, 1995) and PT (Gonzalo and Granger, 1995) of the Mainland China stock market, denoted by *IS_MC* and *PT_MC*, which is the relative ratio (compared with Hong Kong stock market) of its contribution to price discovery for each pair of A-H cross-listed stocks.

We take the Logistic transformation because IS and PT lie between 0 and 1 and it is reasonable to match its range with other explanatory variables.

Since the IS and PT are already relative ratios, IS_MC and PT_MC are not absolute values. Rather, they are Mainland China market's relative contribution to informational price discovery compared with the Hong Kong market. For example, for stock JWT, the IS_MC in the year 2005 is 0.657. This implies that on average Mainland China's A-share market contributed 65.7% of price discovery happening in the A- and H- shares market for JWT in 2005. The average IS_MC for our entire sample A-H pairs across their cointegrated years is 0.632 and the average PT_MC for our entire sample A-H pairs across their cointegrated years is 0.645.

5.2 Explanatory variables

In this section, we discuss the reasons to choose the following explanatory variables and the economic hypothesis behind them.

5.2.1 Bid-Ask spread and its components

We expect IS_MC and PT_MC to relate inversely to the relative transaction cost of Mainland China compared with Hong Kong, because lower trading cost make it easier to incorporate information and have it reflect in equilibrium prices. Harris et al. (2002a) indicates that NYSE's relative contribution to price discovery, compared with regional exchanges, relates negatively to relative spreads. Eun and Sabherwal (2003) analyze price discovery for US-Canada cross-listing Canadian stocks. Similarly, they find that the U.S. share of price discovery relates inversely to its ratio of bid-ask spreads. In this paper, we use effective relative spread to represent the transaction cost in each market. The explanatory variable, $EffRelSprd$, denotes the relative transaction cost of Mainland China market compared with Hong Kong as the ratio of the average effective relative spread for each sample A-share in each sample year to that of the corresponding H-share in the same year. Based on the literature and our economic hypothesis, we expect $EffRelSprd$ to inversely relate to IS_MC .

The dynamics of development and one of the main contributions of market microstructure research have been the components of bid-ask spread and its decomposition. As was mentioned in the literature review section, Lin et al. (1995), Huang and Stoll (1997), Madhavan et al. (1997), and Kraus and Stoll (1972) try to describe the influence of information on security prices from a microstructure perspective. Therefore, we combine the econometric methods of measuring price

discovery with the decomposition of bid-ask spreads, which can be decomposed into an order processing component, an inventory component and an adverse selection component. We expect that there will be a positive relationship between the IS of one market and its corresponding adverse selection component of the bid-ask spread.

Since both the Mainland China and the Hong Kong stock markets are order-driven markets, we use the Lin et al. (1995) method to decompose the effective relative spread of each sample A-share and H-share stock in each sample year into the adverse-selection component and order-processing component. The reason for choosing this decomposition method and excluding the inventory component lies in empirical estimations. George et al. (1991), Hasbrouck (1988), Madhavan and Smidt (1991), and Stoll (1989) suggest that inventory-holding costs appear relatively small. This is especially so for electronic markets lacking formal market makers. Hence, order processing and adverse information costs seem to be the main components of the bid-ask spread in an order-driven market. The explanatory variable *AdvSelComp* is the ratio of the adverse selection component in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs. *OrdProcComp* is the ratio of the order processing component in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs. Since *AdvSelComp* represents Mainland China's information advantage relative to the Hong Kong market, we expect to find a significant positive relationship between *IS_MC* (*PT_MC*) and *AdvSelComp*.

5.2.2 Control variables

Since our sample of the dependent variable varies in both cross-section and time-series dimensions, we use dummy variables to control for these effects.

First, we expect that there might be a difference for *IS_MC* (*PT_MC*) across industries. So we divide the A-H pair companies in our sample into eight categories of industries, according to the Shanghai and Shenzhen Stock Exchanges rules, respectively as *Industrials*, *Consumer*, *Materials*, *Energy*, *Telecommunication*, *Health*, *Utilities* and *Finance*. Therefore, we introduce seven categories of industry dummy variables, as *Industrials*, *Consumer*, *Materials*, *Energy*, *Telecommunication*, *Health* and *Utilities*.

Second, we wish to test the hypothesis that stock exchanges themselves have an effect on the price discovery for cross-listing A- and H- shares. A-shares in Mainland China markets list on either the Shanghai Stock Exchange (SSE) or Shenzhen Stock

Exchange (SZSE). We introduce the indicator variable *SZSE* for cross-listing stocks whose A-shares are listed on SZSE.

Last, for the time-series dimension, we consider two categories of indicator variable. The first category tests for the influence of QFII and QDII policies. We expect *IS_MC* (*PT_MC*) to change after implementation of these two policies. The reason for this hypothesis is that these policies allow financial institutions in Mainland China and abroad to invest in both A- and H- shares markets under certain regulatory restrictions. There has been sufficient deregulation for capital to flow between these two markets to some extent, though not enough to flow freely. Su et al. (2007) use daily closing price data to explore the potential existence of comovements between the stock prices in Mainland China and Hong Kong. Their cointegration test shows that the prices of a substantial number of A shares and H shares have started to cointegrate the launch of the Closer Economic Partnership Arrangement (CEPA) in recent years. The second category of time-series indicator variable is simply year dummies. We expect to see some differences between the relationship of *IS_MC* (*PT_MC*) and year dummies after the year 2006, when the QDII allowed domestic commercial banks in Mainland China to invest in the Hong Kong stock market under admittedly restrictive conditions.

5.3 Regression results and implications

In this section, we discuss the regression results based on previous analysis and explain the implications for our hypothesis. Most of the regression results are consistent with our hypothesis expectations. First, we present the regression results for explanatory variables of most importance, *EffRelSprd* and its decomposition components *AdvSelComp* and *OrdProcComp*.

(Insert Table 7.a and 7.b about here)

From Table 7.a, we observe a negative relationship between *IS_MC* and *EffRelSprd*, at a 10% significance level. This is consistent with our expectation that the relative IS of Mainland China compared with Hong Kong is inversely related to the transaction cost represented by its relative ratio of effective relative spread. After we decompose the effective relative spread and calculate the ratio for Mainland China market respectively for its two components, we find a significant positive relationship between *IS_MC* and *AdvSelComp*, but no relationship between *IS_MC* and *OrdProcComp*. If we run a regression of *IS_MC* only on *AdvSelComp*, the positive

relationship becomes even stronger, at a 1% significance level. Table 7.b shows that there is an even stronger positive relationship between *PT_MC* and *AdvSelComp*, with impressive significance compared with *IS_PC*. However, we find no significant relation between *PT_MC* and *EffRelSprd*. We do find that both *IS_MC* and *PT_MC* significantly and positively relate to the relative ratio of adverse selection component of the effective relative spread of the corresponding market. Since the adverse selection component represents the information advantage of informed traders, the result implies that the reason for Mainland China having a higher IS (PT) than Hong Kong is significantly due to the information advantage of domestic investors in Mainland China A-shares market. It also supports the underlying hypothesis of the adverse selection component from Lin et al. (1995) that it is due to the presence of informed traders. Moreover, we are the first to show definitively that both measures of price discovery, IS and PT, capture informed trading and price discovery. The latter shows a more significant positive relationship with the informed trading indicator.

Other than previously summarized literature on the information advantage related to home bias, several recent articles which particularly investigate this issue for cross-listing stocks and international portfolios include Kalev et al. (2008), Bae et al. (2008), Barron and Ni (2008) and Karlsson and Norden (2007). They all examine the information asymmetry between local and foreign investors and provide some evidence that local investors have trading advantages for cross-listed stocks. Hong et al. (2008) test the local bias using data on U.S. states and it is consistent with a home bias and geographic proximity effect for internationally cross-listing stocks. Not only on the investment side is there a ‘home bias’, Sarkissian and Schill (2004) imply that the same proximity constraints for investment portfolios also exert a profound influence on financing decisions. Kang and Kim (2008) examine the importance of geographic proximity in acquirer and target returns to find that block acquirers have a strong preference for geographically proximate targets. These realize higher announcement returns and better post-acquisition operating performance than do targets of other types of acquirers, particularly when they face greater information asymmetries. Therefore, it is reasonable for us to conclude that information asymmetry is an important reason for Mainland China having a higher IS (PT).

Next, we present the regression results after controlling for cross-section and time-series effects to test the robustness of our results. Table 8.a and Table 8.b demonstrate the relations between the dependent variable and explanatory variables after

controlling for, respectively, industry and policy dummies, stock exchange and policy dummies, and stock exchange and year dummies. To control and test for both cross-sectional and time-series effects, we add different categories of dummy variables. The A-H pair listing companies in our sample can be divided into eight categories of industries, according to the Shanghai and Shenzhen Stock Exchanges, respectively as Industrials, Consumer, Materials, Energy, Telecommunication, Health, Utilities and Finance. Therefore, we introduce seven categories of industry dummy variables, as *Industrials, Consumer, Materials, Energy, Telecommunication, Health, and Utilities*.

For time series dummy variables, we divide the 2001-2007 sample period into three sub-periods according to the QFII and QDII policies adopted by the Chinese government. Implementation of QFII and QDII policies occurs, respectively, during 2002 and 2006. Therefore, we introduce two time dummy variables, respectively *PostQFPreQD* (implying the time-period after the QFII policy in 2002 and before the QDII policy in 2006) and *PostQD* (implying the time-period after the QDII policy). According to the stock exchange on which the A-shares of these cross-listing stocks list, respectively the Shanghai Stock Exchange (SSE) and Shenzhen Stock Exchange (SZSE), we introduce a dummy variable *SZE* for cross-listed stocks that have A-shares listed in SZSE. Last, we divide the entire 2001-2007 sample period into seven sub-period yearly intervals. Hence, we introduce six time dummy variables, *year1, year2, year3, year4, year5, and year6*, respectively, for each year from 2001 to 2006.

(Insert Table 8.a and 8.b about here)

From Table 8.a and Table 8.b, we draw the main conclusions with respect to the more fundamental dummy variables, both in cross-sectional and time-series dimensions. The positive relationship between *IS_MC* (*PT_MC*) and *AdvSelComp* is still significant in all the three regressions. In Table 8.a, the first regression shows that, for cross-listing companies in certain industries, such as *Industrials, Consumer, Materials, Telecommunication* and *Health*, the IS for Mainland China A-shares market is higher than that of Hong Kong H-shares market. By contrast, Table 8.b shows that only for companies in the *Health* industry that the PT in Mainland China is significantly higher than that of Hong Kong. In the second regression of both Tables 8.a and Table 8.b, we test for the hypothesis that different stock exchanges will have a different effect on price discovery. The results for IS (Table 8.a) strongly support the hypothesis that cross-listing companies, whose A-shares are listed in the Shenzhen Stock Exchange (SZSE), have a significantly higher IS than the Hong Kong market,

compared with companies listed in Shanghai Stock Exchange (SSE). It indicates that, for A-shares listed in SZSE, the domestic investors in Mainland China have a greater information advantage than A-shares listed in SSE, compared with H-shares investors in Hong Kong market. This advantage is not significant for regression result using the PT method (See Table 8.b). From the third regression of IS (Table 8.a), we find that after the year 2006, the IS for Mainland China decreased significantly at the 5% level. In both regression 1 and 2, we do not see any strong relationship between QFII (QDII) policy dummies and IS_MC. Hence, it is reasonable to believe that only QDII (carried out in 2006) has a strong effect on the price discovery for cross-listing A- and H-shares. By contrast, from the third regression of PT (Table 8.b), we find that for several year dummies, including 2002, 2004, 2005 and 2006, there are significant negative relationship between them and the PT of Mainland China, indicating a gradual increase of Hong Kong's share of price discovery in terms of PT method.

In summary for the regression results from Table 7.a, Table 7.b, Table 8.a, and Table 8.b, we find strong evidence that Mainland China market has a higher IS or PT, because domestic investors in Mainland China have an information advantage and geographic proximity compared with foreign investors in the Hong Kong stock market. Policy changes instigated by Mainland China, especially opening-up the capital market, have also had an influence on the domestic investors' position of informational advantage.

6. Conclusions

In this paper, we examine the price discovery process for cross-listing A- and H-shares, the shares of a same company simultaneously listed in the Mainland China and the Hong Kong stock market. Our study includes a sample of 30 pairs of A-H shares through 7 years from 2001 to 2007, with 26 A-shares listed in Shanghai Stock Exchange and 4 A-shares listed in Shenzhen Stock Exchange in Mainland China stock market. The sample data include intraday data of two hours of overlapping trading time on every trading day for both A- and H-shares markets.

Based on a cointegration model, we use the Hasbrouck (1995) IS method and the Gonzalo and Granger (1995) PT method to discuss which market contributes more to the price discovery process for cross-listing A- and H- shares. Our year-by-year cointegration test, from 2001 to 2007, shows that the prices for Mainland China A-shares market and Hong Kong H-shares market are becoming more and more

cointegrated through these seven years, as a consequence of the relative growth in either the number or volume of Hong Kong trades relative to the Mainland. This provides one of the first systematic tests providing an explanation for cointegration of cross-listed stocks and the development of Mainland China as an emerging market.

Our empirical results for the IS method provide strong evidence that the Mainland China market plays a predominant role in price discovery for cross listing A- and H-shares, because 97.4% sample pairs show that the A-shares market has a higher IS than the H-shares market. Results of the PT method does not show a formidable dominance for Mainland China market in the price discovery process, but still 72.7% sample pairs indicate that Mainland China market's share of price discovery is higher than that of the Hong Kong market. Consequently, it is reasonable to draw the conclusion that Mainland China market dominates the price discovery for most cross-listing A- and H- shares during their cointegrated years.

To explore the reason why A-shares market dominates most of the price discovery for cross-listing A- and H- shares, we ran regressions of IS (PT) for Mainland China on different explanatory and control variables, which have an economic meaning and are of great interest to policy makers. Our most original contribution is to combine the econometric methods such as IS and PT with the decomposition of bid-ask spreads from the market microstructure literature, to show that they not only complement but also provide cross-validation of each other. The regression results are consistent with our expectation. Most importantly, there is a significantly positive relationship between the relative IS (PT) of Mainland China and its relative adverse selection component of the effective relative spread, with the PT showing more significance. Because the adverse selection component represents the information advantage of informed traders, the result implies that the reason for Mainland China having a higher IS (PT) than Hong Kong is significantly due to the information advantage and geography proximity of domestic investors in Mainland China A-shares market.

The regression results for controlling dummy variable both in cross-section and time-series dimensions show that, for cross-listing companies whose A-shares are listed in the Shenzhen Stock Exchange (SZSE), they have a significantly higher IS than the Hong Kong market, compared with companies listed in Shanghai Stock Exchange (SSE). In addition, after the year 2006, the relative level of IS for Mainland China significantly drops compared with previous years, which indicates that the

QDII policy is having a strong effect on the investment behavior in the A- and H-shares market with Hong Kong traders becoming more informationally rich.

We are also concerned with further integrating the use of econometric methods in price discovery with key contributions from the market microstructure framework. The influence of the drastic roller coaster changes in A-shares market in the last couple of years is also of great interest. These both represent useful directions for future research.

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Table 1: Literature summary and comparison for price discovery in cross-listing countries

Literature	Cross-listing countries	Methodology	Main results
Werner and Kleidon (1996)	U.K. and U.S.	Intraday pattern	Domination varies
Domowitz et al. (1998)	Mexico and U.S.	Liquidity and volatility	Trade-off impact
Hupperets et al. (2002)	Holland and U.S.	Hasbrouck (1995)	Domination varies
Eun and Sabherwal (2003)	Canada and U.S.	Hasbrouck modified	Domination varies
Wang and Jiang (2003)	Chinese mainland and HK	Daily data regression	Domination varies
Biais and Martinez (2004)	Germany and France	Information content	Home dominates
Gammig et al. (2005)	Germany and U.S.	Hasbrouck modified	Home dominates
Pascual et al. (2006)	Spain and U.S.	Hasbrouck modified	Home dominates
Halling et al. (2007)	34 countries and U.S.	Comparison	Domination varies
Menkveld et al. (2007)	Holland and U.S.	State space model	Home dominates

Table 2: The sample of 30 pairs of A- and H- shares and their basic information

No.	Company Abbr. name	Symbol	A-share code ^a	A-share IPO date	H-share code ^a	H-share IPO date	First Listing ^b
1	NE ELECTRIC	NEE	000585.sz	1995-12-13	0042.hk	1995-7-6	HK
2	TSINGTAO BREW	TDB	600600.ss	1993-8-27	0168.hk	1993-7-15	HK
3	JIANGSU EXPRESS	JSE	600377.ss	2001-1-16	0177.hk	1997-6-27	HK
4	BEIREN PRINT MACH	BRP	600860.ss	1994-5-6	0187.hk	1993-8-6	HK
5	KUNMING MACHINE	KMM	600806.ss	1994-1-3	0300.hk	1993-12-7	HK
6	GUANGZHOU SHIP	GZS	600685.ss	1993-10-28	0317.hk	1993-8-6	HK
7	MAANSHAN IRON	MASI	600808.ss	1994-1-6	0323.hk	1993-11-3	HK
8	SHANGHAI PEICHEM	SHP	600688.ss	1993-11-8	0338.hk	1993-7-26	HK
9	JINGWEI TEXTILE	JWT	000666.sz	1996-12-10	0350.hk	1996-2-2	HK
10	JIANGXI COPPER	JXC	600362.ss	2002-1-11	0358.hk	1997-6-12	HK
11	SINOPEC CORP	SNP	600028.ss	2001-8-8	0386.hk	2000-10-18	HK
12	SHENZHEN EXPRESS	SZE	600548.ss	2001-12-25	0548.hk	1997-3-12	HK
13	NANJING PANDA	NJP	600775.ss	1996-11-18	0553.hk	1996-5-2	HK
14	CHINA EAST AIR	CEA	600115.ss	1997-11-5	0670.hk	1997-2-5	HK
15	ZTE	ZTE	000063.sz	1997-11-18	0763.hk	2004-12-9	MC
16	GUANGZHOU PHARMA	GZP	600332.ss	2001-2-6	0874.hk	1997-10-30	HK
17	HUANENG POWER	HNP	600011.ss	2001-12-6	0902.hk	1998-3-4	HK
18	ANHUI CONCH	AHC	600585.ss	2002-2-7	0914.hk	1997-10-21	HK
19	HISENSE KELON ELEC	HKE	000921.sz	1999-7-13	0921.hk	1996-7-23	HK
20	ANHUI EXPRESS WAY	AHE	600012.ss	2003-1-7	0995.hk	1996-11-13	HK
21	YIZHENG CHEM	YZC	600871.ss	1995-4-11	1033.hk	1994-3-29	HK
22	CHINA SOUTH AIR	CSA	600029.ss	2003-7-25	1055.hk	1997-7-31	HK
23	TIANJIN CAPITAL	TJC	600874.ss	1995-6-30	1065.hk	1994-5-17	HK
24	HUADIAN POWER	HDP	600027.ss	2005-2-3	1071.hk	1999-6-30	HK
25	DONGFANG ELEC	DFE	600875.ss	1995-10-10	1072.hk	1994-6-6	HK
26	LUOYANG GLASS	LYG	600876.ss	1995-10-31	1108.hk	1994-6-29	HK
27	CHINA SHIP DEV	CSD	600026.ss	2002-5-23	1138.hk	1994-11-11	HK
28	YANZHOU COAL	YZC	600188.ss	1998-7-1	1171.hk	1998-4-1	HK
29	CHINA MERCHANT BANK	CMB	600036.ss	2002-4-9	3968.hk	2006-9-22	MC
30	BANK OF CHINA	BOC	601988.ss	2006-7-5	3988.hk	2006-6-1	HK

^a The A-share code symbol .ss indicates listing in SSE, .sz in SZSE and H-share code symbol .hk indicates listing in HKEx.

^b This column shows the first listing market for each pair, HK stands for Hong Kong and MC stands for Mainland China.

Table 3: Estimates and tests of Cointegration over the sample years

For each cross-listing company, we use the 1- minute-interval, mid-quote time-series intraday prices for its A-share and H-share in each year to estimate the Johansen cointegration test statistics. We present the estimation results of maximum Eigenvalue test of $H_0: r = 0$ against $H_1: r > 0$, where r is the cointegration rank. A rejection of the null hypothesis indicates existence of non zero cointegrating vector between the A- and H-share price time series. The critical value to reject the null hypothesis at the 5% significance level is 15.34. An asterisk indicates rejection of the null hypothesis $H_0: r = 0$. For years through 2001 to 2007, we estimates the years when data are available for sample stocks and calculate the frequency and percentage of significantly cointegrated A-H share pairs for each sample year. We can observe a gradual and obvious increase in the percentage of cointegrated A-H share pairs from the year 2001 to 2007.

No.	Symbol	Year						
		2001	2002	2003	2004	2005	2006	2007
1	NEE	6.8751	16.5155*	11.2973	7.6003	10.9231	19.3589*	20.8489*
2	TTB			13.8998	16.8367*	30.8019*	4.4996	29.0363*
3	JSE			6.1059	16.1464*	10.0497	12.2298	19.3741*
4	BRP			11.391	14.7918	11.563	32.3354*	30.6553*
5	KMM			9.2814	22.967*	54.8307*	15.1142	25.2627*
6	GZS			13.1562	11.2154	10.7432	15.9514*	21.7893*
7	MASI			5.244	6.7216	18.4269*	17.0302*	18.0557*
8	SHP			11.5977	9.8632	25.1459*	18.4985*	21.0191*
9	JWT	8.7429	5.1591	18.1254*	10.6313	24.0619*	22.36*	15.3651*
10	JXC			10.4515	6.7958	28.6784*	24.8979*	21.7916*
11	SNP			8.1376	27.6928*	16.4912*	11.3748	12.0155
12	SZE			9.4036	7.0542	8.925	50.2874*	19.7572*
13	NJP			6.0872	26.3034*	7.1919	23.9139*	31.8525*
14	CEA			16.1654*	16.2152*	24.7038*	8.2298	14.2711
15	ZTE	18.4892*	10.6751	39.7597*	9.3113	16.996*	17.0964*	21.2273*
16	GZP			15.5017*	5.8721	15.8875*	22.1671*	35.886*
17	HNP			4.9749	15.7588*	21.2087*	9.0233	17.4064*
18	AHC			16.8808*	10.4783	28.7881*	40.6569*	23.3268*
19	HKE	10.2663	15.4989*	19.7737*	12.9472	8.247		
20	AHE			5.7613	9.7384	7.5483	17.0149*	18.6485*
21	YZC			6.0706	15.332	13.4722	17.9941*	30.0842*
22	CSA			14.4615	20.1727*	32.6493*	9.024	19.4514*
23	TJC			13.4444	8.1395	13.7186	12.3445	18.647*
24	HDP					22.1283*	6.1637	17.5796*
25	DFE			9.4772	6.1873	15.8093*	16.6046*	19.3863*
26	LYG			8.2787	7.399	10.5287	28.9011*	
27	CSD			4.9379	11.0197	21.6207*	19.6939*	24.0951*
28	YZC			5.8318	8.7279	8.5508	16.3193*	15.7735*
29	CMB						7.0942	14.4349
30	BOC						50.8064*	16.7506*
No. of significant A-H cointegrated pairs		1	2	6	8	16	19	25
No. of total A-H pairs		4	4	27	27	28	29	28
Frequency of significant A-H cointegrated pairs		1/4	2/4	6/27	8/27	16/28	19/29	25/28
Percentage of significant A-H cointegrated pairs		25.00%	50.00%	22.22%	29.63%	57.14%	65.52%	89.29%

Table 4: Logistic regression results for cointegration explanation

The dependent variable is *COINT*, the dummy for whether a given pair of A-H share prices are cointegrated in a given year. If a given pair of A-H share prices are cointegrated in a given year, *COINT*=1, otherwise *COINT*=0. We report the regression results for three models in order. The independent variable in each model is *R_count*, *R_vol_sum* and *R_spread*. They are respectively the ratio of trading frequency of Mainland China to that of Hong Kong, the ratio of total trading volume of Mainland China to that of Hong Kong and the ratio of relative spread of Mainland China to that of Hong Kong, for a given A-H pair in a given year. There are 147 observations used in each Logistic regression. The *Chi-square statistics* are in the parentheses below the coefficients. Significance at the 1, 5, 10 percent levels is indicated by ***, ** and *, respectively.

	(1)	(2)	(3)
Intercept	0.49** (4.26)	0.53** (5.00)	0.24 (0.41)
<i>R_count</i>	-0.06*** (9.99)		
<i>R_vol_sum</i>		-0.35*** (11.30)	
<i>R_spread</i>			-1.28 (0.97)

Table 5: Results of IS for cointegrated A-H pairs

The mid-point of upper and lower bounds of ISs of Mainland China and Hong Kong stock markets, MC stands for Mainland China and HK for Hong Kong. For each A-H pair in each cointegrated year, the ISs of MC and HK sum up to 1, according to our method. Bolded ISs implies Hong Kong's IS higher than 0.5, which means higher than that of Mainland China. We also provide the maximum and minimum values of ISs for each market in each year.

No.	Symbol	Mid-point ISs (IS) for each A-H pair in each cointegrated year													
		2001		2002		2003		2004		2005		2006		2007	
		MC	HK	MC	HK	MC	HK	MC	HK	MC	HK	MC	HK	MC	HK
1	NEE			0.691	0.309							0.680	0.320	0.658	0.342
2	TTB							0.589	0.411	0.531	0.469			0.628	0.372
3	JSE							0.578	0.423					0.613	0.387
4	BRP											0.668	0.332	0.710	0.290
5	KMM							0.724	0.276	0.639	0.362			0.634	0.366
6	GZS											0.587	0.413	0.568	0.432
7	MASI									0.679	0.322	0.546	0.455	0.598	0.402
8	SHP									0.620	0.381	0.553	0.448	0.633	0.368
9	JWT					0.775	0.225			0.657	0.344	0.661	0.339	0.678	0.322
10	JXC									0.604	0.396	0.564	0.436	0.950	0.050
11	SNP							0.605	0.396	0.561	0.440				
12	SZE											0.512	0.489	0.571	0.429
13	NJP							0.541	0.460			0.657	0.344	0.726	0.274
14	CEA					0.652	0.348	0.601	0.400	0.638	0.362				
15	ZTE	0.681	0.319			0.650	0.350			0.650	0.351	0.550	0.450	0.611	0.390
16	GZP					0.568	0.432			0.639	0.361	0.598	0.403	0.678	0.322
17	HNP							0.659	0.342	0.680	0.320			0.575	0.426
18	AHC					0.521	0.479			0.576	0.425	0.585	0.416	0.573	0.428
19	HKE			0.637	0.363	0.730	0.271								
20	AHE											0.562	0.438	0.583	0.417
21	YZC											0.508	0.493	0.524	0.477
22	CSA							0.631	0.370	0.607	0.393			0.623	0.377
23	TJC													0.605	0.396
24	HDP					0.566	0.434			0.636	0.365			0.532	0.468
25	DFE									0.537	0.463	0.558	0.443	0.658	0.342
26	LYG											0.619	0.381		
27	CSD									0.608	0.392	0.509	0.492	0.581	0.420
28	YZC											0.548	0.452	0.563	0.438
29	CMB														
30	BOC											0.480	0.520	0.496	0.504
	Max for each year	0.681	0.319	0.691	0.363	0.775	0.479	0.724	0.460	0.680	0.469	0.680	0.520	0.950	0.504
	Min for each year	0.681	0.319	0.637	0.309	0.521	0.225	0.541	0.276	0.531	0.320	0.480	0.320	0.496	0.050

Table 6: Results of PT method for cointegrated A-H pairs

The following table presents the shares of price discovery of Mainland China and Hong Kong using PT method, for each pair of A-H shares in each cointegrated year. The shares of price discovery of MC and HK sum up to 1, according to our method. Bolded shares of price discovery implies Hong Kong's share higher than 0.5, which means higher than that of Mainland China. We also provide the maximum and minimum values of shares of price discovery under PT method for each market in each year.

No.	Symbol	Shares of price discovery using PT method													
		2001		2002		2003		2004		2005		2006		2007	
		MC	HK	MC	HK	MC	HK	MC	HK	MC	HK	MC	HK	MC	HK
1	NEE			0.748	0.252							0.781	0.219	0.787	0.213
2	TTB							0.684	0.316	0.480	0.520			0.537	0.463
3	JSE							0.616	0.384					0.744	0.256
4	BRP											0.849	0.151	0.724	0.276
5	KMM							0.851	0.149	0.806	0.194			0.942	0.058
6	GZS											0.764	0.236	0.720	0.280
7	MASI									0.443	0.557	0.161	0.839	0.601	0.399
8	SHP									0.470	0.530	0.619	0.381	0.818	0.182
9	JWT					0.948	0.052			0.326	0.674	0.378	0.622	0.753	0.247
10	JXC									0.248	0.752	0.324	0.676	0.724	0.276
11	SNP							0.855	0.145	0.403	0.597				
12	SZE											0.371	0.629	0.762	0.238
13	NJP							0.853	0.147			0.674	0.326	0.618	0.382
14	CEA					0.708	0.292	0.997	0.003	0.776	0.224				
15	ZTE	0.884	0.116			0.988	0.012			0.422	0.578	0.785	0.215	0.742	0.258
16	GZP					0.909	0.091			0.909	0.091	0.890	0.110	0.941	0.059
17	HNP							0.018	0.982	0.262	0.738			0.521	0.479
18	AHC					0.714	0.286			0.363	0.637	0.381	0.619	0.940	0.060
19	HKE			0.703	0.297	0.995	0.005								
20	AHE											0.587	0.413	0.364	0.636
21	YZC											0.507	0.493	0.549	0.451
22	CSA							0.602	0.398	0.381	0.619			0.720	0.280
23	TJC													0.817	0.183
24	HDP									0.281	0.719			0.916	0.084
25	DFE									0.592	0.408	0.947	0.053	0.953	0.047
26	LYG											0.706	0.294		
27	CSD									0.345	0.655	0.995	0.005	0.452	0.548
28	YZC											0.215	0.785	0.510	0.490
29	CMB														
30	BOC											0.434	0.566	0.551	0.449
	Max for each year	0.884	0.116	0.748	0.297	0.995	0.292	0.997	0.982	0.909	0.752	0.995	0.839	0.953	0.636
	Min for each year	0.884	0.116	0.703	0.252	0.708	0.005	0.018	0.003	0.248	0.091	0.161	0.005	0.364	0.047

Table 7.a: Regression results for IS using components of Bid-Ask spread

The dependent variable *IS_MC* is the Logistic transformation of the IS (Hasbrouck 1995) of the A-shares in Mainland China stock market, which is the relative ratio (compared with Hong Kong stock market) of its contribution to price discovery for A-H pair cross-listing stocks.

	(1)	(2)	(3)
Intercept	0.64*** (32.65)	0.60*** (62.93)	0.60*** (65.82)
<i>EffRelSprd</i>	-1.87* (-1.62)		
<i>AdvSelComp</i>		0.10** (2.11)	0.11*** (2.59)
<i>OrdProcComp</i>		0.04 (0.61)	
Adjusted R-square (%)	2.14	5.26	6.07

Table 7.b: Regression results for PT using components of Bid-Ask spread

The dependent variable *PT_MC* is the Logistic transformation of the PT (Gonzalo and Granger 1995) of the A-shares in Mainland China stock market, which is the relative ratio (compared with Hong Kong stock market) of its contribution to price discovery for A-H pair cross-listing stocks.

	(1)	(2)	(3)
Intercept	-3.45 (-0.91)	0.60*** (20.55)	0.61*** (21.63)
<i>EffRelSprd</i>	4.12 (0.28)		
<i>AdvSelComp</i>		0.48*** (3.19)	0.52*** (3.60)
<i>OrdProcComp</i>		0.17 (0.84)	
Adjusted R-square (%)	0.23	13.54	13.89

For both Table 7.a and Table 7.b, *EffRelSprd* is the ratio of the average effective relative spread for each sample A-share in each sample year to that of the corresponding H-share in the same year. We use the Lin et al. (1995) method to decompose the effective relative spread (*EffRelSprd*) of each sample A-share and H-share stock in each sample year and get the Adverse Selection component and the Order Processing component. The explanatory variable *AdvSelComp* is the ratio of the Adverse Selection component in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs in each year. *OrdProcComp* is the ratio of the Order Processing component in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs in each year. There are 76 observations used in this panel data regression, including 29 A-H pair companies across 7 years. The *t-statistics* are in the parentheses below the coefficients. Significance at the 1, 5, 10 percent levels is indicated by ***, ** and * respectively.

Table 8.a: Regression results for IS after adding different categories of dummy variables

The dependent variable *IS_MC* is the Logistic transformation of the IS (Hasbrouck 1995) of the Mainland China stock market, which is the relative ratio (compared with Hong Kong stock market) of its contribution to price discovery for A-H pair cross-listing stocks using the IS method. The explanatory variable *AdvSelComp* is the ratio of the Adverse Selection component (Lin et al. (1995)) in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs in each year. *OrdProcComp* is the ratio of the Order Processing component (Lin et al. (1995)) in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs in each year. We introduce seven categories of industry dummy variables, as *Industrials*, *Consumer*, *Materials*, *Energy*, *Telecommunication*, *Health* and *Utilities*. For time series dummy variables, we introduce two time dummy variables, respectively *PostQFPreQD* (implying time period after the QFII policy in 2002 and before the QDII policy in 2006) and *PostQD* (implying time period after the QDII policy). According to the stock exchange that the A-shares of these cross-listing stocks are listed, we introduce a dummy variable *SZE* for cross-listing stocks whose A-shares are listed in SZSE. Lastly, we divide the 2001-2007 sample period to seven sub-periods according to the year itself. Hence, we introduce 6 time dummy variables, *year1*, *year2*, *year3*, *year4*, *year5* and *year6* respectively for the years from 2001 to 2006. There are 76 observations used in these 3 regressions, including 29 A-H pair companies across 7 years. The 3 regressions are run on 3 different categories of dummy variables presented in Table 8.a. We display the *t-statistics* in the parentheses below the coefficients. Significance at the 1, 5, 10 percent levels is indicated by ***, ** and *, respectively.

	(1)	(2)	(3)
Intercept	0.49*** (6.52)	0.55*** (9.47)	0.62*** (41.87)
<i>AdvSelComp</i>	0.12** (2.07)	0.11** (2.09)	0.13** (2.07)
<i>OrdProcComp</i>	0.04 (0.46)	0.03 (0.38)	-0.03 (-0.27)
<i>Industrials</i>	0.12** (2.36)		
<i>Consumer</i>	0.15*** (2.72)		
<i>Materials</i>	0.12** (2.25)		
<i>Energy</i>	0.10 (1.66)		
<i>Telecommunication</i>	0.13** (2.34)		
<i>Health</i>	0.13** (2.14)		
<i>Utilities</i>	0.11* (1.84)		
<i>PostQFPreQD</i>	-0.01 (-0.25)	0.06 (1.08)	
<i>PostQD</i>	0.02 (0.32)	0.03 (0.53)	
<i>SZE</i>		0.06*** (2.67)	0.06*** (2.50)
<i>year1</i>			0.03 (0.27)
<i>year2</i>			-0.09 (-1.22)
<i>year3</i>			-0.03 (-0.83)
<i>year4</i>			-0.03 (-0.95)
<i>year5</i>			-0.01 (-0.62)
<i>year6</i>			-0.05** (-2.48)
Adjusted R-square (%)	8.09	14.10	14.16

Table 8.b: Regression results for PT after adding different categories of dummy variables

The dependent variable *PT_MC* is the Logistic transformation of Mainland China's shares of price discovery using the Gonzalo and Granger (1995) PT method, which is the relative ratio (compared with Hong Kong stock market) of its contribution to price discovery for A-H pair cross-listing stocks using the PT method. The explanatory variable *AdvSelComp* is the ratio of the Adverse Selection component (Lin et al. (1995)) in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs in each year. *OrdProcComp* is the ratio of the Order Processing component (Lin et al. (1995)) in Mainland China market to that of the Hong Kong market for each corresponding A-H pairs in each year. We introduce seven categories of industry dummy variables, as *Industrials*, *Consumer*, *Materials*, *Energy*, *Telecommunication*, *Health* and *Utilities*. For time series dummy variables, we introduce two time dummy variables, respectively *PostQFPreQD* (implying time period after the QFII policy in 2002 and before the QDII policy in 2006) and *PostQD* (implying time period after the QDII policy). According to the stock exchange that the A-shares of these cross-listing stocks are listed, we introduce a dummy variable *SZE* for cross-listing stocks whose A-shares are listed in SZSE. Lastly, we divide the 2001-2007 sample period to seven sub-periods according to the year itself. Hence, we introduce 6 time dummy variables, *year1*, *year2*, *year3*, *year4*, *year5* and *year6* respectively for the years from 2001 to 2006. There are 76 observations used in these 3 regressions, including 29 A-H pair companies across 7 years. The 3 regressions are run on 3 different categories of dummy variables presented in Table 8.b. We display the *t-statistics* in the parentheses below the coefficients. Significance at the 1, 5, 10 percent levels is indicated by ***, ** and *, respectively.

	(1)	(2)	(3)
Intercept	0.20 (1.03)	0.32* (1.78)	0.70*** (16.29)
<i>AdvSelComp</i>	0.48*** (3.30)	0.62*** (4.02)	0.77*** (3.56)
<i>OrdProcComp</i>	0.41* (1.76)	0.32 (1.34)	0.01 (0.02)
<i>Industrials</i>	0.20 (1.46)		
<i>Consumer</i>	0.06 (0.42)		
<i>Materials</i>	0.03 (0.23)		
<i>Energy</i>	0.03 (0.19)		
<i>Telecommunication</i>	0.19 (1.27)		
<i>Health</i>	0.40** (2.50)		
<i>Utilities</i>	-0.07 (-0.47)		
<i>PostQFPreQD</i>	0.21 (1.45)	0.21 (1.26)	
<i>PostQD</i>	0.37** (2.43)	0.37** (2.13)	
<i>SZE</i>		0.06 (0.88)	0.02 (0.26)
<i>year1</i>			0.16 (0.56)
<i>year2</i>			-0.49** (-2.29)
<i>year3</i>			-0.06 (-0.52)
<i>year4</i>			-0.23** (-2.27)
<i>year5</i>			-0.24*** (-3.62)
<i>year6</i>			-0.13** (-2.05)
Adjusted R-square (%)	37.81	22.32	28.23

Figure 1: The trading hours for Mainland China and Hong Kong stock markets

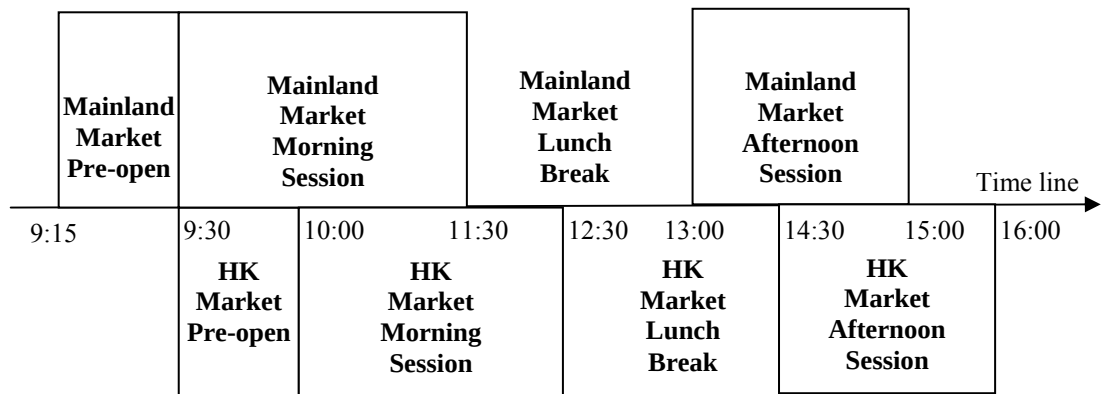
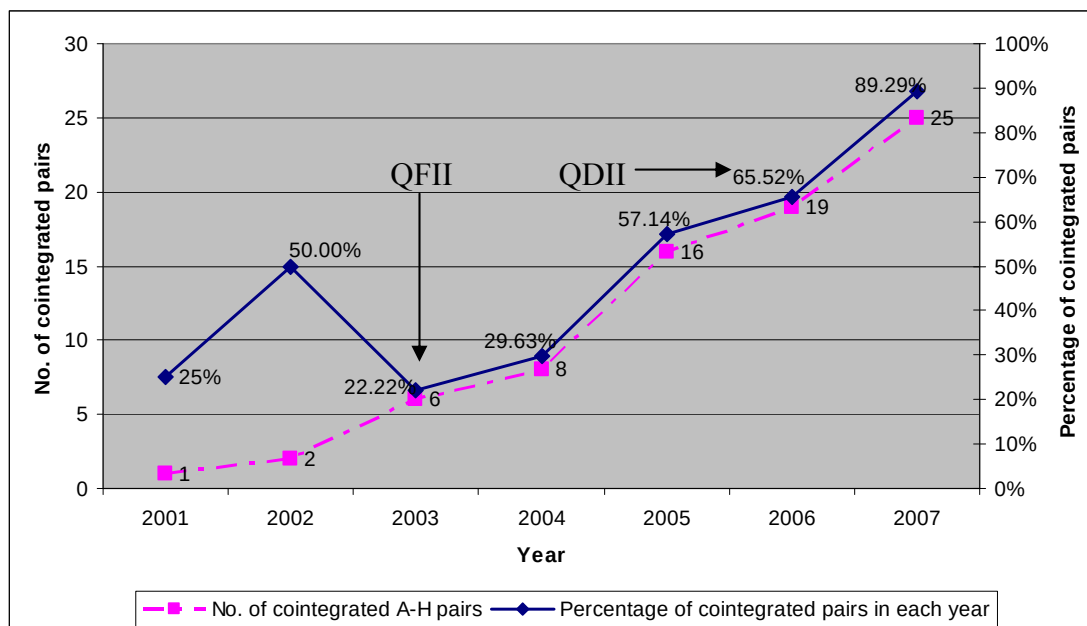
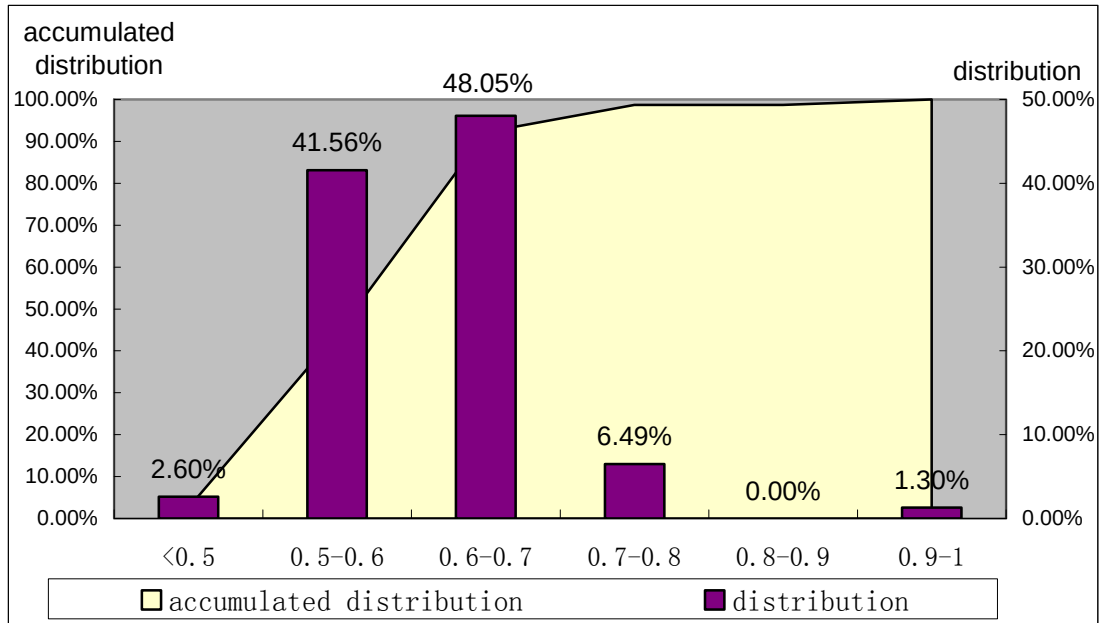


Figure 2: Evolution trend of cointegrated A- and H-shares pairs in each year



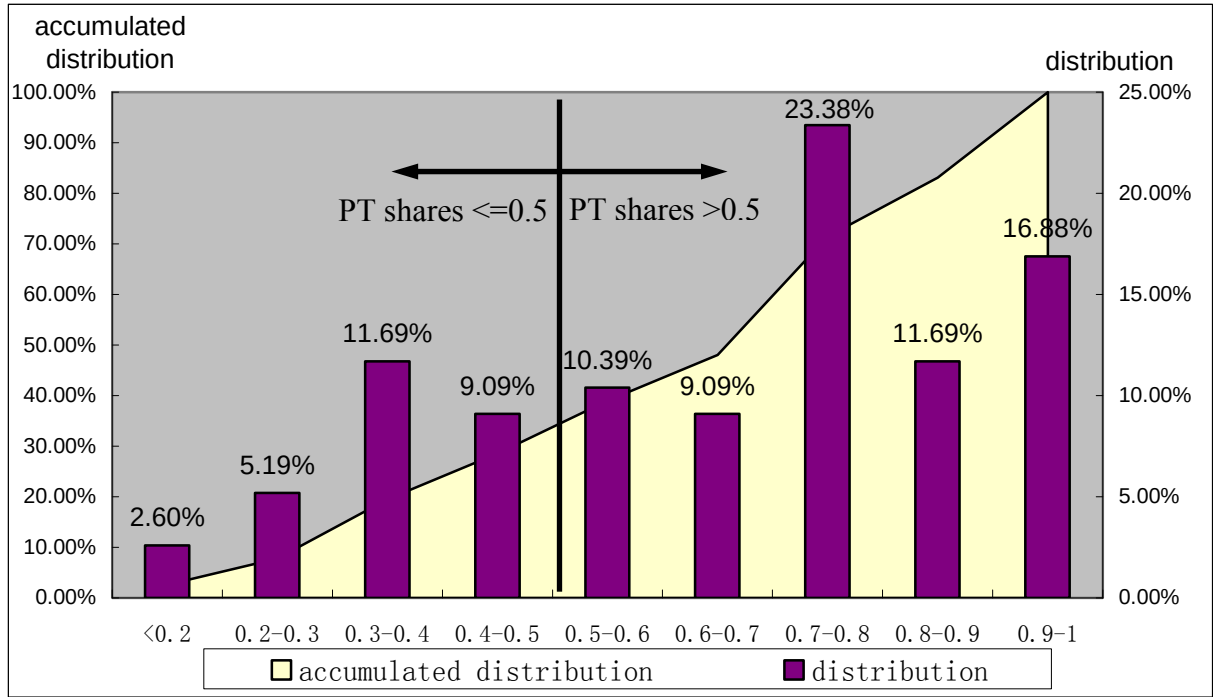
The dashed line represents the absolute number of cointegrated A-H pairs in each year from 2001 to 2007, while the solid line shows the percentage of cointegrated A-H pairs out of all available samples in each year.

Figure 3: The distribution of Information Shares for Mainland China



To illustrate Table 5, in this figure, we present the distribution and accumulated distribution of Information Shares for Mainland China in all the cointegrated years of our sample A- and H- shares, according to the estimation of Table 3. The dark bars show the distribution of ISs for Mainland China and the light area shows the accumulated distribution of ISs for Mainland China, which respectively fall in the categories of “ $IS < 0.5$ ”, “ $0.5 < IS \leq 0.6$ ”, “ $0.6 < IS \leq 0.7$ ”, “ $0.7 < IS \leq 0.8$ ”, “ $0.8 < IS \leq 0.9$ ” and “ $0.9 < IS \leq 1$ ”. Since we normalize the ISs such that the Information Shares for Mainland China and Hong Kong sum up to 1, an IS higher than 0.5 for Mainland China indicates a dominant role for Mainland China market in price discovery.

Figure 4: The distribution of shares of price discovery for Mainland China (PT method)



To illustrate Table 6, in this figure, we present the distribution and accumulated distribution of shares of price discovery for Mainland China using the PT method in all the cointegrated years of our sample A- and H- shares, according to the estimation of Table 3. The dark bars show the distribution of shares of price discovery for Mainland China and the light area shows the accumulated distribution of shares of price discovery for Mainland China, which respectively fall in the categories of “ $PT < 0.2$ ”, “ $0.2 < PT \leq 0.3$ ”, “ $0.3 < PT \leq 0.4$ ”, “ $0.4 < PT \leq 0.5$ ”, “ $0.5 < PT \leq 0.6$ ”, “ $0.6 < PT \leq 0.7$ ”, “ $0.7 < PT \leq 0.8$ ”, “ $0.8 < PT \leq 0.9$ ” and “ $0.9 < PT \leq 1$ ”. Since we normalize the shares of price discovery under the PT method such that the shares of price discovery for Mainland China and Hong Kong sum up to 1, a “PT” share of price discovery higher than 0.5 for Mainland China indicates a leading role for Mainland China market in price discovery.